Honest Fit/Gap Report: Case Study Report

Candidate	Ahmad Rian
Position	Frontend Engineer Intern
Pull request	github.com/rakamindev/ai-interview-platform/pull/204
Video	Walkthrough (3:56), Indonesian narration with English captions
Branch	`ahmad-rian/ai-interview-platform`, `feat/honest-fit-gap-report` (head `a3f51fb`)
Date	28 October 2026

TL;DR

1. **The product made uncertainty look certain.** A skill nobody asked about was stored as "L1, Confidence: High" and printed as "Gap -2" in the PDF a hiring manager receives; that PDF crashed for any portfolio a human had corrected.
2. **The human check was not durable.** Overrides vanished on regeneration, needed no reason, could not be undone and never showed in fit/gap. Another organisation could read and overwrite a candidate's ratings.
3. **The revamp makes the flow say only what the evidence supports**: "not assessed" is a real state, low confidence never counts toward fit, every human decision has a note and a history, fit/gap is computed on read, and the PDF is rendered from the same presenters as the screen.
4. **It was built test first** on a repository that had no tests and no CI: 228 RSpec examples and 251 Vitest tests, every fix red by test name before green, and three seeded faults caught by CI and reverted in visible history.
5. **What remains is out of the revamp's scope, not hidden**: candidate consent and Indonesian UI, the bandwidth gate, production boot configuration and Gemini capacity are written up as follow-ups with owners.
6. **Frontend focus.** Most of the visible work is frontend: a design system and visual foundation for every assessor page (colour tokens tuned for contrast, `PageShell` and `PageHeader`, `ListRow`, `StatusBadge`), the Fit/Gap and Portfolio screens, the override, confirm, undo and history dialogs, and the Needs review section. Accessibility is part of the design: everything works from the keyboard, every review action has a unique name per skill, text meets WCAG AA contrast, touch targets are 44 px, and pages fit 360 px without horizontal scrolling. The visual audit found the LevelRadio data bug (the level label of skill 3 changed skill 1), fixed with a red test first, and the web suite has 251 tests. The backend side (tenant isolation, the PDF and the shared presenters) was built as well.

Step 2: Context and domain

Insights

Insight	Why it matters for the revamp
The decision artefact is the PDF, not the screen (PRD-02 ends with an export for the hiring manager)	The PDF must carry the same states as the screen: not assessed, low confidence, overrides, AI label
Indonesian youth unemployment is 17.37% (May 2026) and about 2.55 million graduates enter the market each year	When screening filters volume, one fabricated gap silently removes a person
Structured interviews are the strongest selection method, but LLM judges are inconsistent and overconfident	Value comes from evidence, anchors and calibrated human review, not automation
UU PDP 27/2022 Pasal 10 gives candidates the right to object to decisions based solely on automated processing; PP 33/2026 (in force 16 January 2027) requires a human alternative	Human review has to be real, recorded and visible, not a free-text field
Tenant isolation was broken (finding C1)	In production this is a breach with a 3 x 24 hour notification duty (Pasal 46)

Who is affected

User	Need	Harm before the revamp
Candidate	Be judged only on what was asked and answered	Unmeasured skills counted as gaps; data visible to other organisations; no notice or consent
Assessor	Correct the AI with a reason that sticks	Corrections lost on regeneration; no history; PDF crash
Recruiter, hiring manager	One honest document per candidate and vacancy	Empty Required column, hidden overrides, template text that read like analysis, endless spinner

UU PDP mapping (consent, purpose, Pasal 10, DPIA under Pasal 34, cross-border transfer under Pasal 56) is in `02-context-and-domain.md`, sections 6 and 7.

Step 3: Problems and gap to the ideal

Core problem

The product turns uncertain interview evidence into output that looks certain, and the human check that should catch it is neither durable nor visible. On top of that, candidate records were not isolated between organisations.

Findings (37: P0 4, P1 19, P2 11, P3 3)

ID	Sev	Finding	Impact (one sentence)	Status
A1	P0	Missing or invalid AI level coerced to L1 to L5	A skill never rated becomes "Gap -2", so the candidate is marked down for something never measured	Fixed (slice 2)
B1	P0	Regeneration deletes assessor overrides	An assessor's correction disappears and the candidate silently reverts to the AI rating	Fixed (slice 3)
C1	P0	Cross-organisation read and write	Another company can export and overwrite a candidate's ratings	Fixed (slice 1)
C2	P0	No candidate notice or consent	Candidates are recorded and scored by AI without the information UU PDP requires	Out of scope (Q5), memo
A2	P1	Model output not schema-validated	One malformed field fails the portfolio or silently changes values	Fixed (slice 2)
A3	P1	Confidence ignored in fit/gap; AI confidence shown next to human level	A gap backed by one weak quote looks identical to a proven gap	Fixed (slices 2, 4, 6a)
A4	P1	"Not assessed" missing from totals	Two unmeasured skills read as "Match: 3, 0 gaps"	Fixed (slices 4, 6a)
A5	P1	Narrative provenance not shown	Template text on timeout reads like an AI assessment	Fixed (slice 4, D-8)
A6	P1	Stale fit/gap after portfolio changes	Old levels returned as current	Fixed by design (D-3)
A8	P1	Level label in skill 3 changed skill 1	The wrong required level reached fit/gap	Fixed (slice 7)
A9	P1	Failed load showed an empty, saveable edit form	Saving erased the vacancy's culture and competency text	Fixed (slice 7)
B2	P1	PDF export crashed for overridden portfolios	The official document could not be produced after a human correction	Fixed (slice 5)
B3	P1	Override invisible in fit/gap	Human review could not be shown or audited	Fixed (slices 4, 6a)
B4	P1	Override needed no note, no undo, history overwritten	No record of why a level changed	Fixed (slice 3)
B5	P1	PDF without decision states	The official document overstated certainty	Fixed (slice 5)

ID	Sev	Finding	Impact (one sentence)	Status
D1, D2	P1	Polling re-enqueued jobs; report had no status	11 duplicate jobs observed and an endless spinner	Fixed (slices 4, 6a)
D3	P1	Portfolio written without a transaction	A mid-write failure left a partial portfolio	Fixed (slice 2)
F1	P1	Required column empty (seam: `expected_level` vs `required_level`)	Levels shown with nothing to compare against	Fixed (slices 4, 6a)
E1, E2, G1, G2	P1	Bandwidth gate, invite host, production boot, Gemini capacity	Candidates blocked or links wrong; ops risk	Memo items, out of scope
B6, D4, D5, D6, F2	P2	Reports deleted by overrides, failure without reason, export errors, race, mobile overflow	Lost data, invisible failure, broken mobile	Fixed (slices 2 to 8)
C3	P2	Raw internal errors sent to client and logs	Leaky API and logs	Fixed in the touched paths (codes and error classes only); session start/end handlers still log `e.message`
A7, C4, E3, E4, G3	P2	Taxonomy id dropped, third-party calls, English-only, candidate 401 redirect, FK on delete	Various	Out of scope, listed
F3, F4, G4	P3	"2" vs "L2", empty "(expected)", docs drift	Inconsistency	F3 and "(expected)" fixed; tenant label and docs remain

Missing specification vs broken implementation

Eight findings are pure gaps in the wiki specification (human review trail, narrative provenance, report lifecycle, candidate notice) and two more are both a gap and a defect (A1, A3); they needed product decisions, which were taken as Q1 to Q5. The rest are implementation defects against the wiki (for example PRD-01 section 5 defines confidence as a rule, the code asked the model).

API and web seam

The same data was computed in one layer and interpreted differently in another: `expected_level` vs `required_level`, `is_override` never emitted, confidence absent from the web type, `not_assessed` missing from totals, 404 meaning both "not generated" and "generating". Root cause: no shared, typed contract and two renderers (React and Prawn). Decision D-5 introduced one presenter per concept for JSON, web and PDF.

Constraint signal to the Tech Lead

Escalated before building: tenant isolation (blocker), production boot configuration, the LLM output contract, the "not assessed" semantics change to fit/gap totals, report lifecycle schema changes, Gemini capacity and configuration, the human-in-the-loop policy, candidate-side compliance and data retention. Full memo: `03-problems-and-gaps.md`, section 7.

Step 4: Options, trade-offs and acceptance criteria

ID	Chosen	Rejected and why
D-1 Not assessed	Nullable level + `rating_status` + reason; "no evidence, no level"	Level 0 sentinel enters arithmetic and recreates the bug
D-2 Overrides	Stable skill key, upsert, append-only events, note required, undo, 409 on conflict	Portfolio versioning is too invasive for the deadline
D-3 Fit/gap	Computed on read; only the narrative is async with status, digest and an atomic claim	A cached report needs every write path to remember invalidation
D-4 Tenancy	Explicit scope through the session's tenant; 404 for other organisations	Denormalised `tenant_id` is the durable follow-up; Pundit needs a new dependency
D-5 Contract	One presenter per concept shared by JSON, web and PDF	Renaming fields one by one lets drift return
D-6 PDF	Bundled Noto fonts, render from presenters, transliterate unsupported glyphs	HTML to PDF adds a system dependency
D-7 Confidence	Computed from the coverage map as the wiki defines	Model self-confidence is overconfident
D-8 Narrative	No template fallback; system summary plus a labelled optional AI narrative	A labelled fallback still impersonates analysis

Priorities. Must: C1, A1, A2, B1, B4, D3, A3 (data), A4, F1, B3, D1, D2, A6, B6, B2. Should: A3 (UI), A5, B5, C3. Could: F2, F3, D4, D5, D6, and slice 8 (Needs review).

Acceptance criteria (66 in total, `04-strategy-and-acceptance-criteria.md`), for example:

AC	Given / When / Then (shortened)
AC-02	Given model output without a level, when the portfolio is generated, then the skill is not assessed with reason `no_level_from_model`, never L1
AC-41	Given an unreviewed low-confidence skill, when fit/gap is computed, then it is `low_confidence`, does not count in either direction and appears under follow-up
AC-60	Given a portfolio of organisation A, when organisation B calls any portfolio, fit/gap, export or override endpoint, then the response is 404 with no writes and no jobs
AC-95	Given a narrative in progress, when the page polls, then it only sends GET and stops on complete, failed or outdated
AC-104	Given the candidate list, when it is requested, then each row has `portfolio_status` and `review_summary`, so a failed portfolio never looks completed

Step 5: What was built

Slices

Slice	Commits	Result
0 Harness	`a1a0de5..98aa8d8`	RSpec, Vitest, CI, Zeitwerk fix, dummy seed
1 Tenant isolation	`1670e37..4a2245a`	Every touched endpoint scoped; 404 for other organisations
2 Not assessed	`84a82b8..21fb11f`	M1, M2; validated output; transactional, claimed generation
3 Overrides	`4905602..9d34050`	M3; durable overrides, required notes, history, undo, confirm
4 Fit/gap on read	`73a2045..6d49aad`	M4; narrative lifecycle; polling only reads
5 PDF	`c5d7f9e..e14354a`	PDF from shared presenters, bundled fonts, text-level tests
6a Fit/Gap UI	`6d64bad..5219efd`	Evidence, coverage and review state; design system; AA colours
6b Portfolio UI	`1b9c73f..951a237`	Review dialogs, 409 handling, results menu
7 Visual foundation	`aa3a55c..38d2f1f`	Two data bugs found in the audit (A8, A9); page shell; 44 px targets; contrast
8 Needs review	`05832a5..a428ad6`	Seven review reasons, review queue API, honest candidate statuses
Final a11y	`ffda702..c19c111`	Review actions and level group named per skill
Login page	`b9954bf..a3f51fb`	Product name, helper text, password toggle

Test evidence

Measure	Before	After
RSpec examples	0 (no `spec/` directory)	228, 0 failures (also with `CI=true`)
Vitest tests	0 (no runner)	251 in 43 files
CI	none	GitHub Actions for API and web on every push and PR; latest run 40369090527 green
Type check, build	not run	`tsc --noEmit` clean, `vite build` succeeds

- Pattern: a `test:` commit that is red by test name on the old code, then the `fix:` or `feat:` commit that makes it green. Each commit was built and checked in a temporary worktree; when a new test file failed only at import, it was also run against stubs so every assertion was seen failing.
- Old assertions were changed only twice, each in its own commit with the old value, the new value and the reason (the "Completed" status that locked in N15, and the review action names for W1).

Seeded fault test

Three faults were planted in a scratch branch, one commit each, then reverted with `git revert`. The branch is named `fix/seeded-fault-demo` because the CI workflow only runs for `feat/`, `fix/` and pull requests; the workflow was not changed.

Fault	Decision it attacks	CI run (fault)	Caught by	CI run (revert)
Missing or invalid levels clamped to 1..5 again	A1, D-1	40310894123 failure	11 `rating_normalizer_spec` examples (AC-02 to AC-04)	40311115824 success

Fault	Decision it attacks	CI run (fault)	Caught by	CI run (revert)
Tenant filter removed from portfolio lookups	C1, D-4	40311288476 failure	4 `tenant_isolation_spec` examples (export, intruder vacancy)	40311417393 success
Unreviewed low confidence counted toward fit	Q1, D-3	40311585888 failure	4 `report_presenter_spec` + 1 `pdf_generator_spec` examples	40311767855 success

AI verification moments (selected)

#	What the AI produced	How it was caught	Evidence
1	Tenant specs that used the victim's vacancy, so the real attack was not tested	First red run contradicted the Step 1 reproduction	`1670e37`, `4a2245a`
2	A claim window based on retries that never happen for POST	Read the `faraday-retry` source before writing the constant	`ec906bb`, finding N16
3	A "green" PDF run that still executed the old generator	Stack trace named a method that only existed in the old file	`c5d7f9e`, `e14354a`
4	`aria-level=4` that passed in jsdom but not in Chrome	Chrome accessibility snapshot showed no level 4	`a584b22`, `d7be01e`
5	A reproduction test that passed on the broken code	"4 failed, 1 passed" in a commit meant to be fully red	`b1522ce`, `68d79e0`

Details: `ai-verification-selected.md`; all 70 moments: `log-bukti.md`.

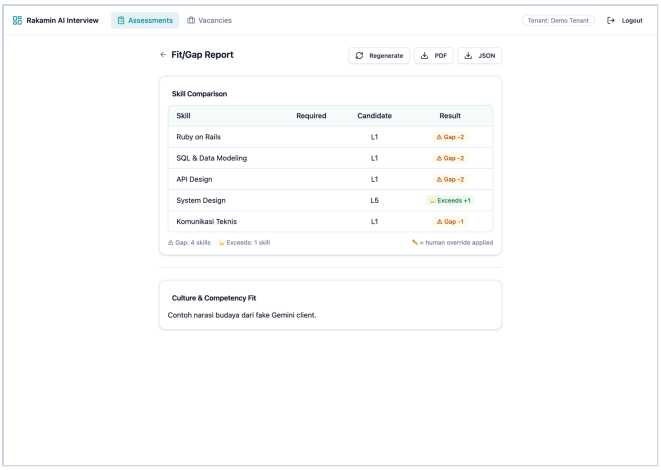
Migrations

Migration	Change	Rollback
M1	Not assessed skills, skill key, coverage snapshot	Refused if not-assessed skills exist (would invent a level)
M2	Legacy flag, safe error code, claim timestamp	Reversible
M3	Override events, lock version, required note	Refused if overrides on not-assessed skills exist
M4	Narrative lifecycle on fit/gap reports	Reversible

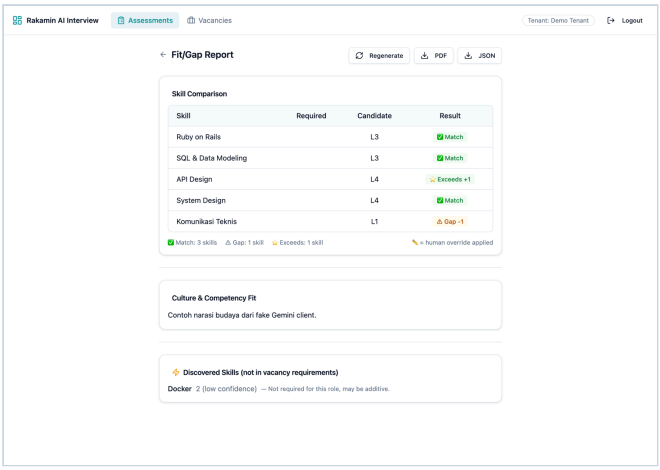
Verified on the test database: rollback of all four to the original schema and migrate again; migrate and seed on an empty database.

Screenshots

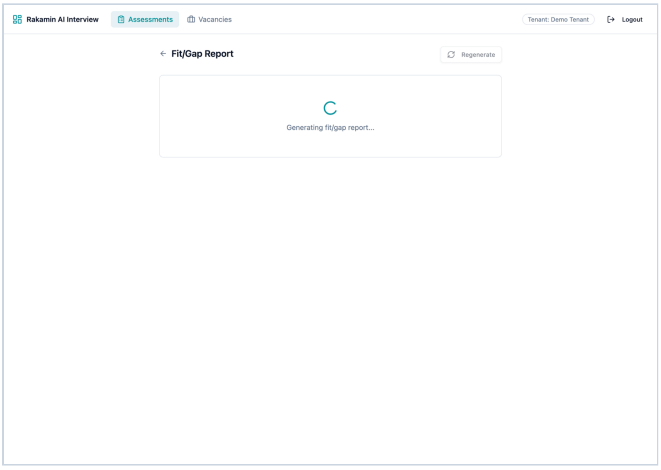
Before



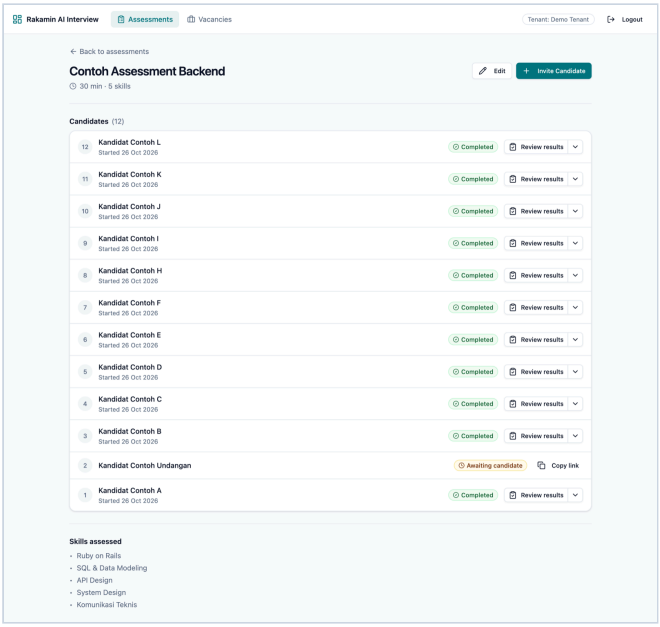
1. Before: a skill the AI never rated shows as L1 and counts as a gap.



2. Before: the Required column is empty and the human override is invisible.

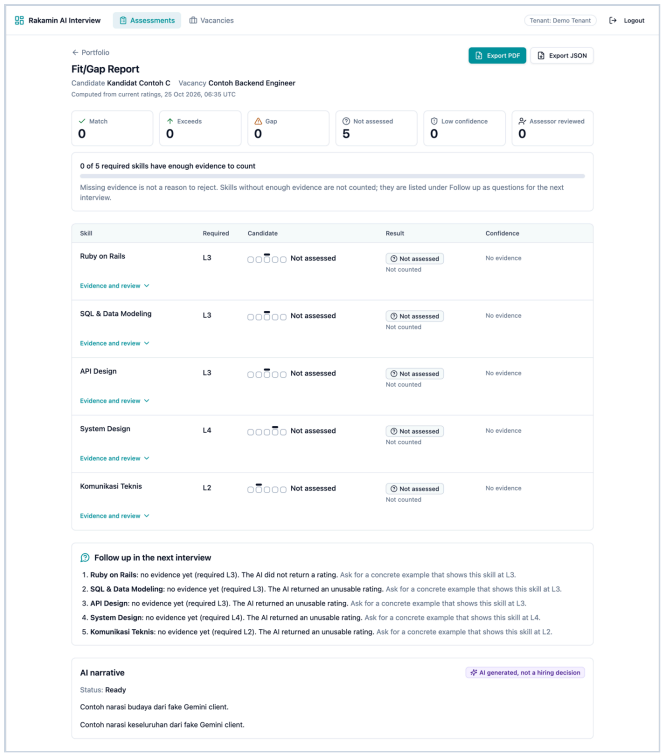


3. Before: a failed report spins forever while a new job is queued every 5 seconds.

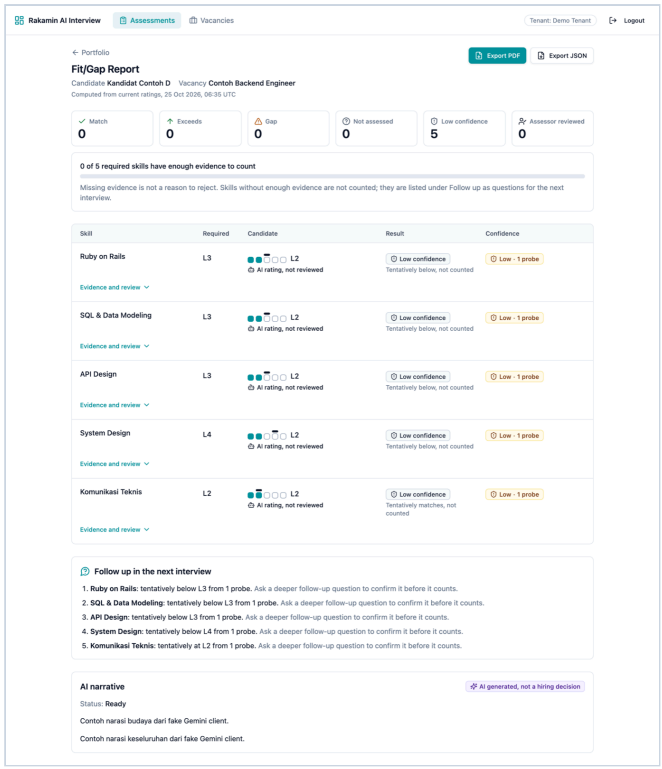


4. Before: every finished interview says "Completed", including failed portfolios (N15).

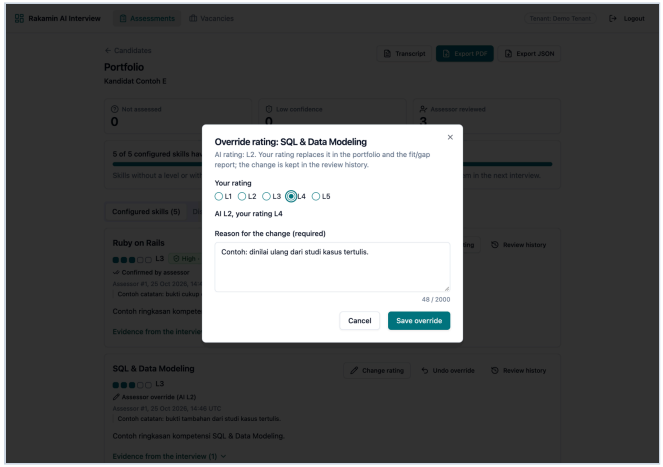
After: the flow



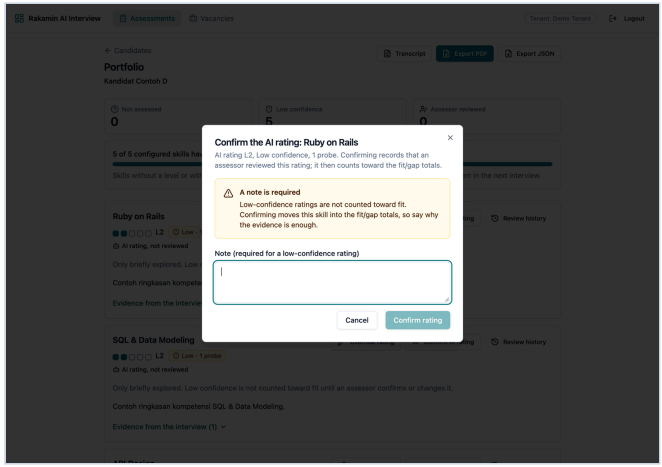
5. Not assessed skills are shown as such, never counted, and listed for follow-up.



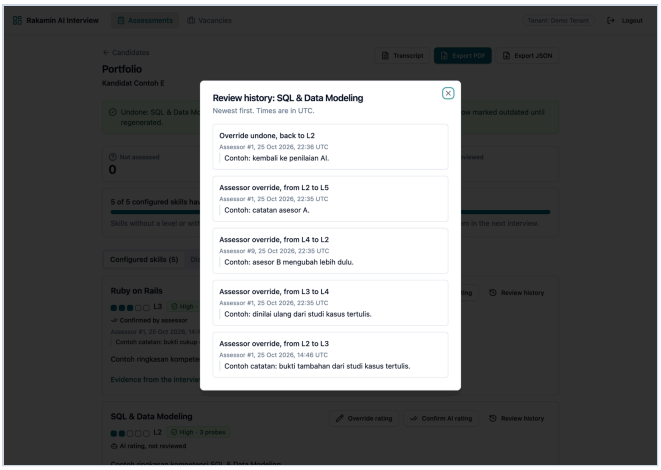
6. Low confidence shows its probe count and a tentative direction, and does not count toward fit.



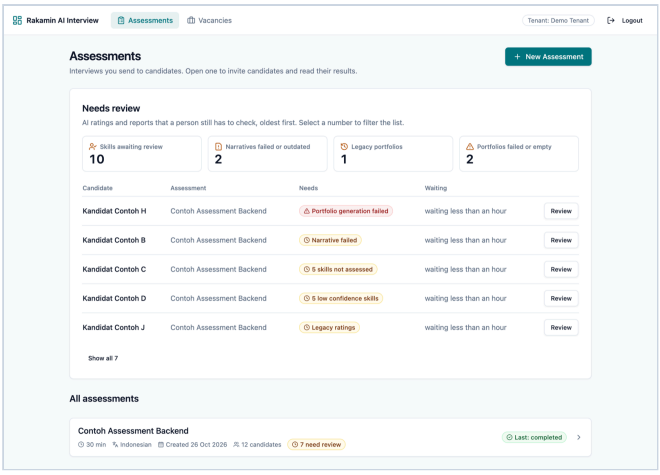
7. Overriding a rating requires a level different from the current one and a reason.



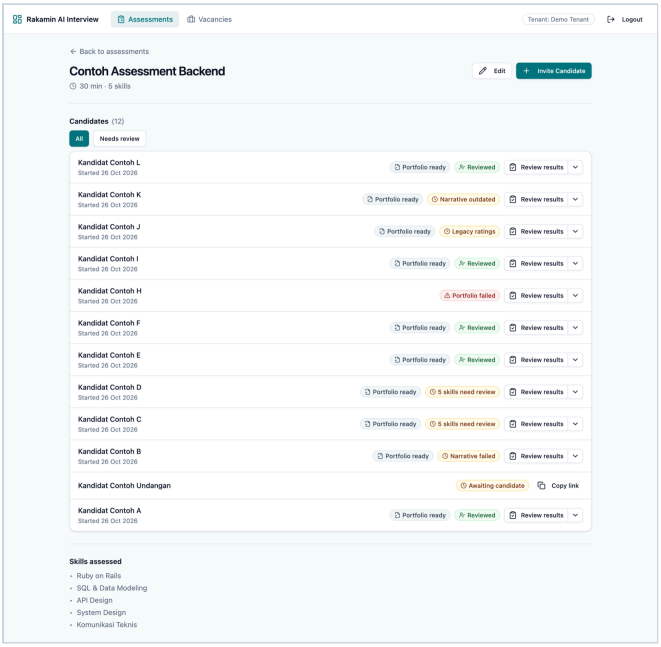
8. Confirming a low-confidence rating requires a note, because it moves the skill into the totals.



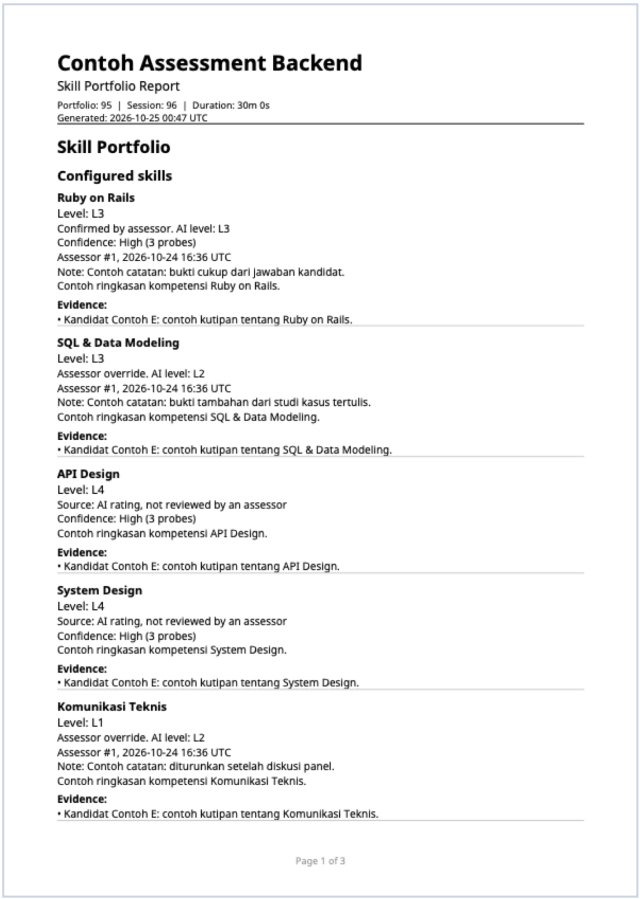
9. Review history lists every set, confirm and undo with author, time (UTC) and note.



10. The assessments page shows what still needs a human, oldest first, with a link to the right page.



11. The candidate list says what each portfolio is and what still needs review.



Contoh Assessment Backend

Skill Portfolio Report

Portfolio: 95 | Session: 96 | Duration: 30m 0s
Generated: 2026-10-25 00:47 UTC

Skill Portfolio

Configured skills

Ruby on Rails
Level: L3
Confirmed by assessor. AI level: L3
Confidence: High (3 probes)
Assessor #1, 2026-10-24 16:36 UTC
Note: Contoh catatan: bukti cukup dari jawaban kandidat.
Contoh ringkasan kompetensi Ruby on Rails.

Evidence:
• Kandidat Contoh E: contoh kutipan tentang Ruby on Rails.

SQL & Data Modeling
Level: L3
Assessor override. AI level: L2
Assessor #1, 2026-10-24 16:36 UTC
Note: Contoh catatan: bukti tambahan dari studi kasus tertulis.
Contoh ringkasan kompetensi SQL & Data Modeling.

Evidence:
• Kandidat Contoh E: contoh kutipan tentang SQL & Data Modeling.

API Design
Level: L4
Source: AI rating, not reviewed by an assessor
Confidence: High (3 probes)
Contoh ringkasan kompetensi API Design.

Evidence:
• Kandidat Contoh E: contoh kutipan tentang API Design.

System Design
Level: L4
Source: AI rating, not reviewed by an assessor
Confidence: High (3 probes)
Contoh ringkasan kompetensi System Design.

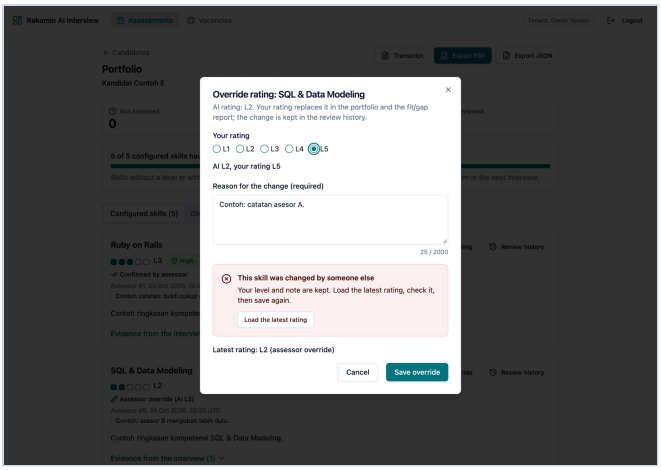
Evidence:
• Kandidat Contoh E: contoh kutipan tentang System Design.

Komunikasi Teknis
Level: L1
Assessor override. AI level: L2
Assessor #1, 2026-10-24 16:36 UTC
Note: Contoh catatan: diturunkan setelah diskusi panel.
Contoh ringkasan kompetensi Komunikasi Teknis.

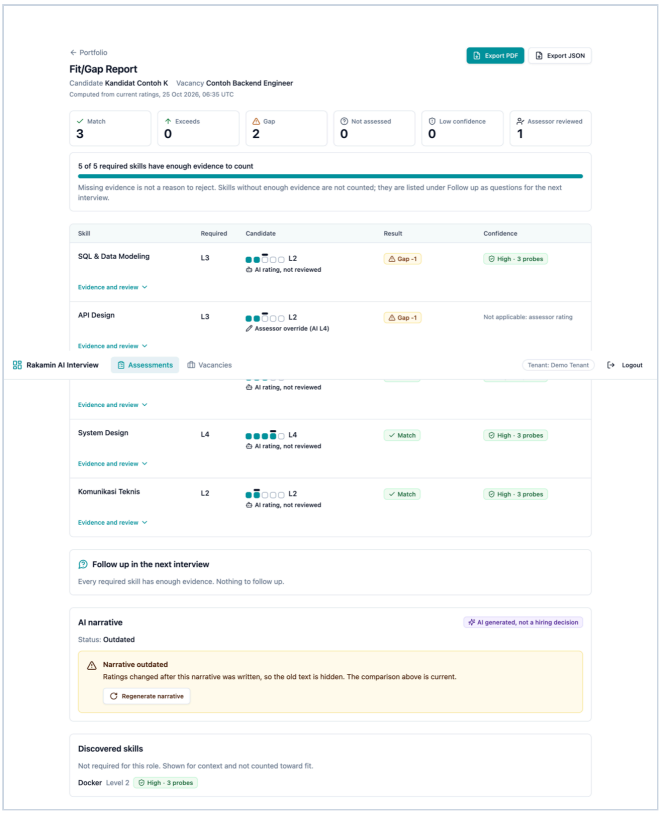
Evidence:
• Kandidat Contoh E: contoh kutipan tentang Komunikasi Teknis.

12. The exported PDF shows the same states as the screen, including overrides and confirmations.

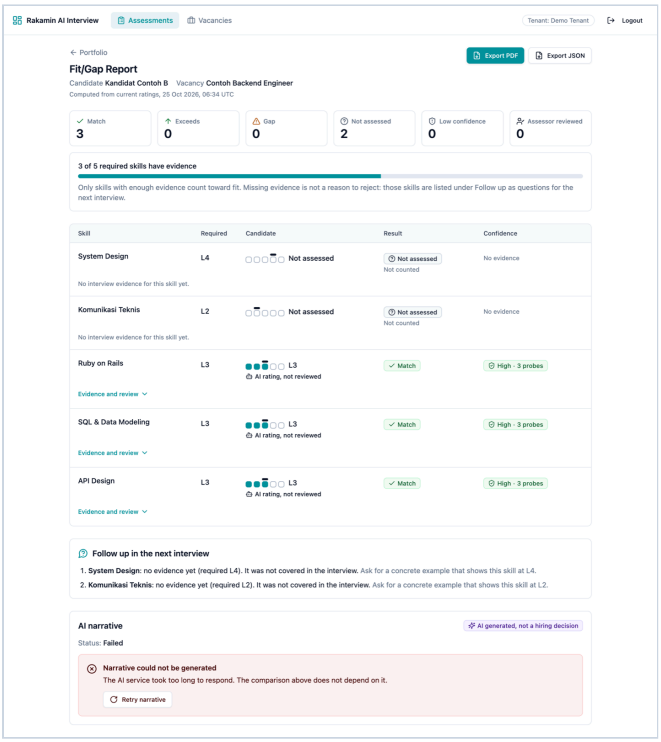
Edge cases and error states



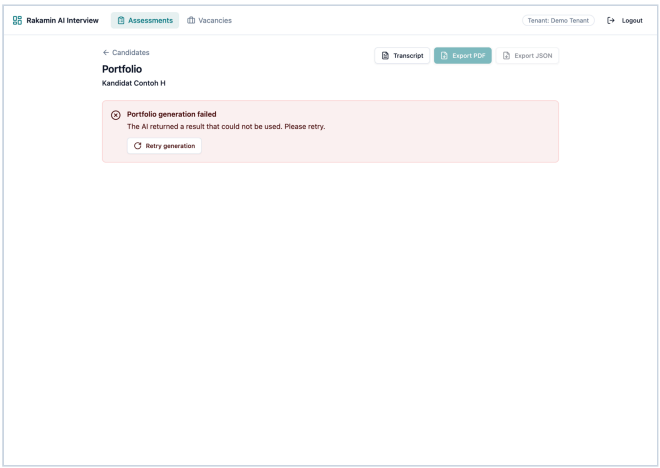
13. A conflicting edit returns 409; the dialog keeps the input and offers to load the latest rating.



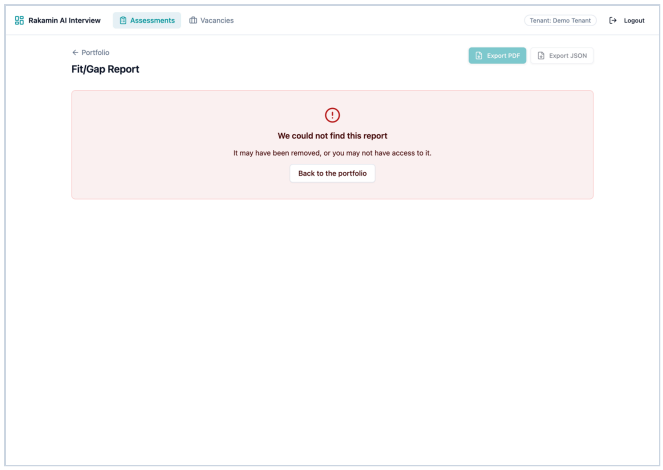
14. A narrative written before a rating changed is marked outdated, not shown as current.



15. When the AI narrative fails, the comparison still stands and the reason and Retry are shown.

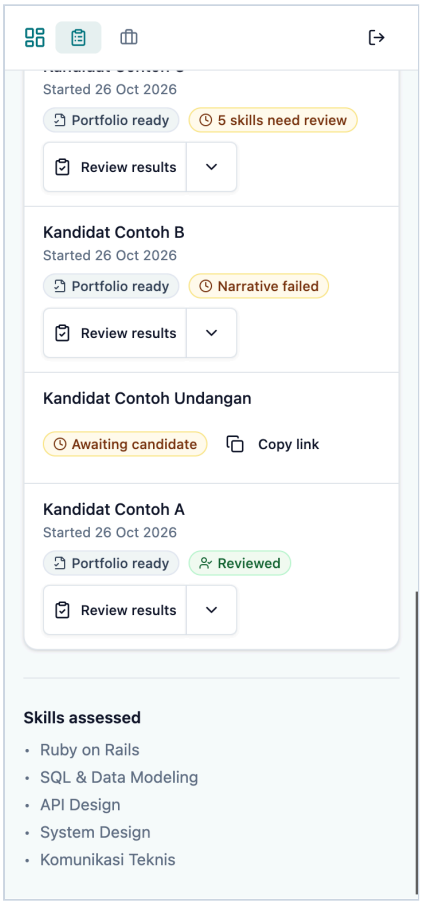


16. A failed portfolio explains why in plain language and offers Retry.

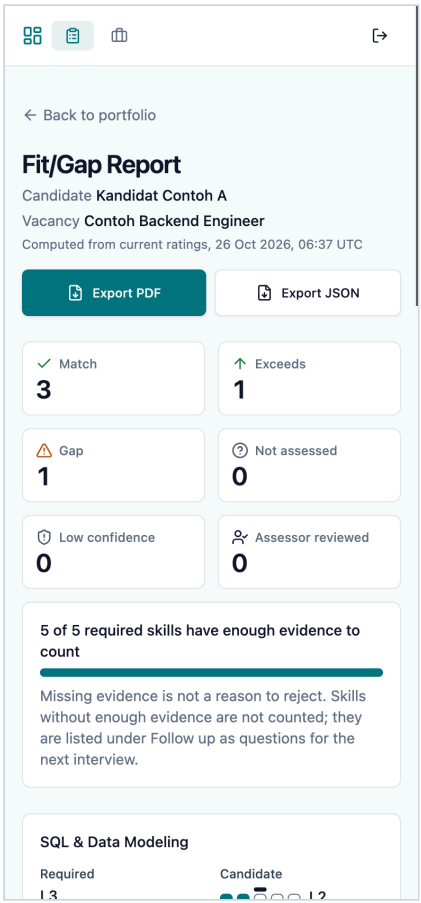


17. Another organisation's portfolio behaves as if it does not exist.

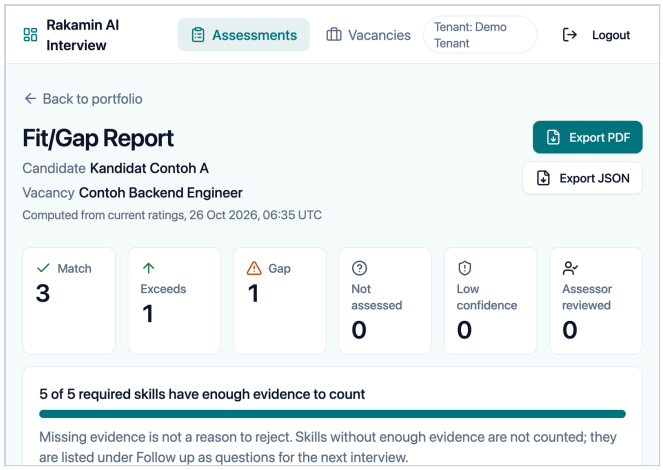
Responsive and accessibility



18. At 360 px the candidate list stacks, with 44 px touch targets and no horizontal scroll.



19. At 390 px the fit/gap report uses cards instead of a table.



20. At 200% text size the summary labels wrap by words instead of being cut off.

Assumptions

Assumption	Where it matters
Hiring managers decide from the PDF and a short conversation, not from the web app	PDF parity was treated as Must
Data volumes are small enough for in-place backfills and for computing review reasons in Ruby	M1 to M4; review queue (9 queries for 50 portfolios)
No external consumers of the fit/gap JSON exist	The contract was changed (`schema_version: 2` in the export)
Gemini acts as a sub-processor and data leaves Indonesia	Memo items on consent and cross-border transfer
Voice is personal data but not biometric unless used for identification	UU PDP mapping
Personas (assessor, HR lead, candidate) come from the wiki and industry context, not user interviews	Step 2
A legacy L1 cannot be told apart from a real L1	Legacy portfolios are flagged, not rewritten
Configured skills are the requirement; discovered skills are extra findings	Review reasons and summaries count configured skills only
"Waiting since" is when the portfolio was generated (or the interview ended for a failed one), because portfolios have no timestamps	Review queue order

Limitations and follow-ups

Item	Status
Live voice interview was not run end to end (microphone, Gemini Live)	Judged from PRD-02, code and seed
Gemini narratives time out or are unavailable on the current quota	The UI shows the reason and Retry; capacity is an ops decision (G2)
Candidate notice, consent, Indonesian UI, realistic bandwidth gate	Out of scope by decision Q5; memo item 8
Denormalised `tenant_id` on portfolio tables	Follow-up to D-4
Production boot configuration (`config.active_job`)	Needs a deploy-file decision (G1)
Deleting a session with a fit/gap report (foreign key)	Open (N7)

Item	Status
Touch target tests check classes; sizes were measured in Chrome	jsdom has no layout
Screen reader use verified through Chrome's accessibility tree, not with VoiceOver or NVDA end to end	Recommended before release
200% text zoom tested with Chrome root font size, not Firefox "zoom text only"	Recommended check